

Lost Spirits: An Interactive Encounter with a Fractured Soul

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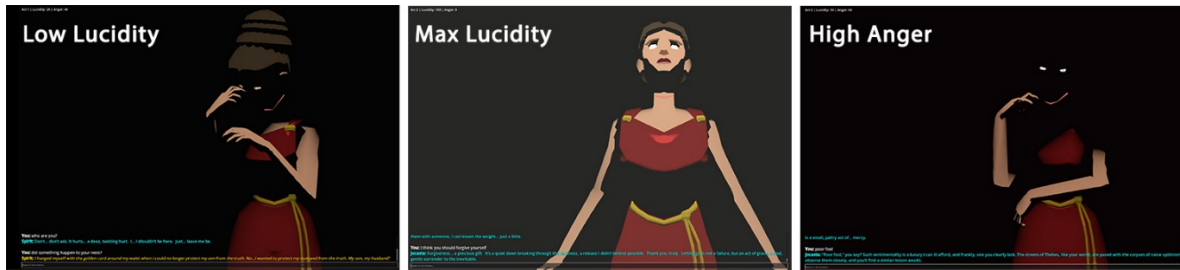


Figure 1: Procedural Performance States. The *Lost Spirits* interface showing dynamic pose shifts triggered by the Affective State Machine. As player input modulates internal meters (Lucidity and Anger), the system triggers corresponding visual states: (Left) Low Lucidity results in a distraught pose; (Center) High Lucidity triggers redemption; (Right) High Anger manifests a contemptuous pose.

Abstract

Lost Spirits is a game prototype developed in the Godot engine, utilizing Gemini Pro, a Large Language Model (LLM) for rapid code generation and a local Small Language Model (SLM), Gemma 3 4B, to power AI-driven NPCs (non-player characters). The study investigates whether LLMs can democratize game development for non-technical artists and if real-time AI-driven character interaction can sustain a meaningful narrative arc without traditional branching scripts. Preliminary results suggest that AI significantly accelerates the technical pipeline—enabling the development of a functional 3D prototype within a two-month window—and facilitates rich, responsive character interactions.

CCS Concepts

• **Computing methodologies**; • **Procedural animation**; **Natural Language Processing**; • **Human-centered computing**; • **Interactive systems and tools**;

Keywords

Human-AI interaction, Edge AI, Local SLMs, Emergent Narrative

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1 INTRODUCTION

The intersection of Generative AI and game design offers two transformative promises: the democratization of game development tools and the creation of truly non-linear storytelling that shifts from traditional NPCs with pre-written branching scripts to AI-driven “Sentient Actors” that generate real-time, unscripted dialogue. The project evaluates if a meaningful narrative arc can be sustained through “Improvisational Alchemy”—a collaboration between the artist/director, the AI/actor, and the player/audience.

This work is inspired and informed by recent industry endeavors into this topic, such as NVIDIA’s ACE framework [NVIDIA, 2025], Meaning Machine’s AI NPC studies [Meaning Machine, 2026] and Google’s “Mystery of the Three Bots” [Google, 2023]. It differs in their approaches however, by focusing specifically on using accessible, affordable AI technologies and game tools for solo artist-developers in the creation of emergent narrative experiences.

2 METHODOLOGY

The project employed a collaborative development pipeline between artist and AI to bridge the gap between a creative idea and the technical skills needed to realize it. In the current prototype of *Lost Spirits*, the player acts as a medium tasked with helping a single fractured ghost reconstruct its identity. As the player uncovers memories, the spirit evolves. The final story arc is a co-authored experience that adapts to the player’s choices and actions.

2.1 Co-Engineering with LLMs

The author utilized Gemini Pro as a real-time coding collaborator to implement game logic in Godot, an open-source game engine. This

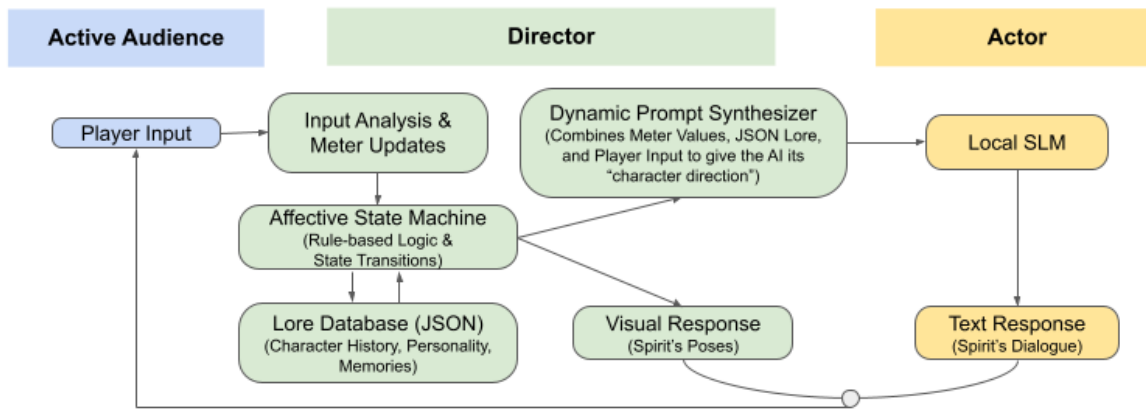


Figure 2: System Architecture of the Affective State Machine. This logic loop facilitates “Improvisational Alchemy” by positioning the developer as Director, the SLM as Actor, and the player as active audience influencing the narrative flow.

workflow functioned as an interactive technical bridge, allowing a solo artist-developer to transition from concept to early-stage prototype within a two-month development window (February–April 2026). By leveraging AI for rapid iteration, complex features—such as JSON parsing and a state machine—were implemented without prior experience in GDScript (Godot’s programming language), or the Godot engine itself.

2.2 Visual Design

To facilitate rapid iteration on consumer hardware, a stylized aesthetic and pose-to-pose animation framework was adopted. The SLM triggers both visual animation poses and dynamically generated dialogue to communicate emotional changes based on its interactions with the player (Figure 1).

2.3 The Improvisational Alchemy Loop

The narrative logic is modeled after improvisational theater, assigning specific roles to the human and AI components:

- 1) **The Artist as Director:** The developer defines the “stage” and “script boundaries” through code and structured JSON data.
- 2) **The SLM as Actor:** Rather than following a static script, the local Gemma model performs the role within the constraints of the Affective State Machine and Dynamic Prompting.
- 3) **The Player as Active Audience:** The player’s natural language input functions as the “audience suggestion,” triggering real-time shifts in the AI actor’s performance and state.

2.4 The Affective State Machine

Lost Spirits utilizes an Affective State Machine to achieve a dynamic, responsive character (Figure 2).

- 1) **Input Analysis:** A trigger-detection system intercepts player input to identify specific “Memory Shards.”
- 2) **State Updates:** Detected keywords dynamically adjust the Lucidity (cognitive coherence) and Anger (emotional valence) meters, which determine the active personality profile.

- 3) **Lore Retrieval:** As Lucidity increases, the system unlocks backstory fragments from a structured JSON database, simulating the ghost “remembering” her past.

2.5 Dynamic Prompt Synthesis

The core of the interaction loop is a Dynamic Prompt Synthesizer. This module assembles a system prompt in real-time by combining the agent’s base persona, the current values of the Affective Meters, and the newly unlocked memory shards. This ensures that the SLM’s output is consistently conditioned by the current emotional context of the game, like a Director’s note given to an actor to guide the performance.

2.6 Edge Computing, Privacy, and Accessibility

The game prototype uses on-device inference via a Godot Engine add-on by developer NobodyWho. This approach removes reliance on expensive cloud APIs and recurring usage fees, significantly lowering the financial barrier for solo creators, and allows the game to run entirely offline. Furthermore, local inference aligns with Responsible AI principles by ensuring that player interactions remain private on the user’s hardware.

3 PRELIMINARY RESULTS

A pilot study of 5 playtesters running an M4 Macbook Air validated the prototype’s ability to generate a responsive, emergent performance on consumer hardware. 80% of participants reported that their input had a “High Impact” on the spirit’s emotional state, with 100% encountering unique, unscripted interactions.

4 CONCLUSIONS AND FUTURE WORK

In this study, the “Co-Engineering” model proved highly effective for democratizing game development, significantly accelerating the acquisition of technical skills within an unfamiliar engine. With the help of AI, an early-stage prototype was developed within two months.

Future research will explore the integration of Retrieval-Augmented Generation (RAG) to scale the narrative beyond a single character to a wider cast of spirits, while maintaining long-term narrative consistency. Additionally, future work will benchmark larger 7B–14B parameter models and different architectures (Gemma vs Nemotron) to find the optimal balance between dramatic nuance and local performance. By demonstrating how AI can empower non-technical creators while remaining affordable and safeguarding user privacy, this project exemplifies a human-centric approach to innovation.

Much uncertainty revolves around AI and its proper use. Given the results of this prototype, however, the author has hope for

successful future collaborations between humans and AI in creating richer, more emergent gaming experiences.

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